CM5244 Topics in Environmental Chemistry (Semester 2, 2016/2017)

Test (12 April 2017)

Examiner: Asst. Prof. Boon Siang Jason YEO

Time allowed: 1 hour

Matric No:_____

INSTRUCTIONS

1. Please write your student number only. **Do not write your name**.

2. There are 6 questions. Answer any 5 of them. If you decide to do all the 6 questions, only questions No. 1-5 will be marked.

3. Write your answers in the space provided below each question.

Question 1

(a) Please describe the mechanistic model for the natural formation and destruction of stratospheric ozone.
(b) Write equations for ozone destruction involving NO• catalyst.
(c) The origin of NO• is thought to be from N_2O, an inert gas that could diffuse from the ground to the stratosphere. Suggest equations for the formation of NO• in the stratosphere.
(d) Suggest potential sinks for removing NO• from the stratosphere.

(5 marks)

(a) The Chapman cycle:

$O_2 + UV \rightarrow 2O$ *(step 1)*
$O + O_2 \rightarrow O_3 + heat$ *(step 2)*
$O_3 + (240\ nm < radiation < 310\ nm) \rightarrow O_2 + O$ *(step 3)*
$O_3 + O \rightarrow 2\ O_2$ *(step 4)*

(b) NO can destroy the ozone via: (The type 1 mechanism in the notes)

$NO + O_3 \rightarrow NOO + O_2$
$NOO + O \rightarrow NO + O_2$
Overall reaction: $O_3 + O \rightarrow 2O_2$

(c) Microbial activity in the soil (from the breakdown of N-containing fertilizers) is thought to be the largest source of N_2O. N_2O is inert, and can diffuse into the stratosphere, where it undergoes:

$N_2O + h\nu \rightarrow N_2 + O$
$N_2O + O \rightarrow 2NO$

(d) For removal: NO can oxidize to form nitrogen dioxide, which then react with water, ultimately forming nitric acid (removal by rain)

$4NO + 3O_2 + 2H_2O \rightarrow 4HNO_3$

N_2O can also react with an O radical to give $N_2 + O_2$.

$N_2O + O \rightarrow N_2 + O_2$

Question 2

'Placed in the stratosphere, CFC-12 has more ozone depleting capabilities than HFC-245'

'HFC-245 exerts a larger greenhouse effect to earth than CFC-12'

Explain how the two statements above can be considered correct.

(5 marks)

CFC-12 = CF_2Cl_2
HFC-245 = $C_3H_3F_5$

Important that you know how to use the 'rule of 90'!

CF_2Cl_2*, when photolyzed, gave 2 Cl radicals which have ozone depleting capabilities, unlike F radicals from* $C_3H_3F_5$*.*

$C_3H_3F_5$ *is a bigger molecule, and has more IR active vibrational modes than* CF_2Cl_2 *(27 vs. 9 modes). Hence, on a per molecule basis,* $C_3H_3F_5$ *should exert a larger radiative forcing (contributes more to global warming) compared to* CF_2Cl_2*.*

Question 3

Describe the role of an automotive catalytic converter in environmental protection. Include the relevant chemical reactions.

(5 marks)

*The **role** of the catalytic converter is to convert harmful emissions such as CO, NO and hydrocarbons from automobiles, into less harmful emissions.*
(NO, in particular, is an important ingredient for the formation of photochemical smog.)

Reduction chamber:

- *NO reduced to N_2 by H_2, which is generated at the Rh catalyst by action of H_2O on unburned fuels.*
  Hydrocarbons + $H_2O \rightarrow H_2 + CO$
  $2NO + 2H_2 \rightarrow N_2 + 2H_2O$
  ($2NO + 2CO \rightarrow N_2 + 2CO_2$ also occurs)

Oxidation chamber:

- *On the surface of a Pt/Pd catalyst:*
  $2CO + O_2 \rightarrow 2CO_2$
  Hydrocarbons + $2O_2 \rightarrow CO_2 + 2H_2O$

Important: It is critical that we operate the three-way catalytic converter at the appropriate air/fuel ratio of 14.7. If the condition is too oxidizing (too much air), the NO will not be able to reduce properly (as CO is consumed). If there is too little air, CO and hydrocarbons cannot be oxidized to CO_2. The temperature of the catalytic converter should also be sufficiently high so that CO can be oxidized to CO_2 on the Pt catalysts.

Question 4

Briefly described how you would *detect and quantify* carbon monoxide (CO) and nitrogen oxides (NO_2, NO) in the air.

(5 marks)

CO: Infrared spectroscopy or gas chromatography.
NO: chemiluminescence.

You should write brief notes of how these techniques work (answers in the lecture notes 2b).

Question 5

In recent years, there has been a drive to incorporate bio-ethanol in diesel. The inclusion of oxygenates (molecules with oxygen atoms) in diesel is suppose to aid in the better oxidation of CO and hydrocarbons to CO_2. Yet concurrently, the use of such fuel mixtures for motor vehicles has been suspected to be a cause of photochemical smogs. Explain the possible underlying cause(s). Include the relevant chemical equations in your answer.

(5 marks)

The answers are in the lecture notes 3, Slides 10-17.

Ethanol tend to oxidize to acetaldehyde. Acetaldehyde can further react to gives peroxyacetyl nitrate (PAN). (Pg 14-15 of lecture notes 3).

Bio-ethanol also tends to contain more N-containing compounds (because of the bio-origin of the ethanol). When combusted, these liberate NO_x.

Question 6

The use of fossil fuels for our energy needs has resulted in undesirable atmospheric pollutions. For global sustainability, *new ways of producing clean electricity and fuels* have to be developed. Describe some of these (please give the underlying science/chemistry of your choices).

(5 marks)

Renewable electricity can be produced from turbines driven by wind, hydro (examples, dams) and geothermal sources. Solar panels (photovoltaics) can also be used.

The electricity that is produced can be stored (converted) in the form of fuels via electrochemical water splitting to produce H_2 gas, or CO_2 reduction to methanol. We could also make fuels from biomass. Details are in the notes. ***Whatever the choice is, it is important that the energy input must be green and sustainable (this is most critical!).***

You should elaborate your answers.

A common error: A fuel cell does not produce fuels (a common misconception). It converts fuels into electricity.

CM5244 Topics in Environmental Chemistry (Semester 2, 2017/2018)

Test (11 April 2018)

Examiner: Asst. Prof. Boon Siang Jason YEO

Time allowed: 1 hour

Student No:_____

INSTRUCTIONS

1. Please write your student number only. **Do not write your name**.
2. There are 5 questions. Answer all 5 of them.
3. Write your answers in the space provided below each question.

Question 1

CFC-211 was formerly used as a refrigerant. With the aid of chemical equations, explain how its release from the ground can adversely affect the concentration of stratospheric ozone. Suggest a method to measure the concentration of CFC-211 in the atmosphere.

(5 marks)

Using the rule of 90, CFC-211 = C_3FCl_7

C_3FCl_7 *is very stable and has no tropospheric sink. Hence, it is able to diffuse to the stratosphere intact. There, it undergoes photolysis to give Cl radicals.*

Cl can destroy the ozone via:

(The type I mechanism in the notes)

$Cl + O_3 \rightarrow ClO + O_2$
$ClO + O \rightarrow Cl + O_2$
Overall reaction: $O_3 + O \rightarrow 2O_2$

(The type II mechanism in the notes)

$Cl + O_3 \rightarrow ClO + O_2$
$Cl + O_3 \rightarrow ClO + O_2$
$ClO + ClO \rightarrow ClOOCl \rightarrow 2Cl + O_2$
Overall reaction: $2O_3 \rightarrow 3O_2$

Having electronegative elements, CFC-211 is suitably quantified using a gas chromatograph with an electron capture detector.

Question 2

When released into the troposphere, CFC-211 can exert a larger *greenhouse effect than water and CO_2.*

I recall reading the above statement in an old magazine! Explain why it **can be considered correct**. If you think it is *fake news*, please justify your thoughts!

(5 marks)

CFC-211 or C_3FCl_7 is a greenhouse gas. It has 3(11)—6=27 vibrational modes. It is also less symmetrical compared to H_2O and CO_2. Intuitively, we can already conclude from here that it has more infrared active vibrations, and that it absorbs more IR compared to H_2O and CO_2. This molecule also absorbs IR in the atmospheric window. Hence, by per molecule basis, CFC-211 exerts a larger radiative forcing compared to H_2O and CO_2, i.e., it exerts a larger greenhouse effect.

Please take note that there is, though, a higher concentration of H_2O and CO_2 in the atmosphere; and hence overall, H_2O and CO_2 remain the major greenhouse gases in the atmosphere.

This question is not about ozone destruction although the molecule in question is a CFC.

Question 3

Biodiesel, which contains methyl esters, is derived from vegetable oils or animal fats. Manufacturers promoting its usage in automobiles, have claimed that it may outperform ethanol, gasoline and conventional diesel in overall fuel-efficiency. Its combustion has also been reported to result in low particulate matter and SO_x emissions. Interestingly, nitrogen oxides (NO_x) emissions are higher. Briefly described, with the aid of chemical equations, how these pollutants can react under stagnant, warm and sunny conditions to give a **photochemical smog**.

(5 marks)

The answers are in the lecture notes 3, Slides 9-15.

Question 4

You have been tasked to **detect** and **quantify** the concentrations of the air pollutants mentioned in Q3. How would you do this?

(5 marks)

PM: BAM counter
NO_x: chemiluminescence.
SO_x: UV-fluorescence

To quantify the concentrations of the pollutants, Beer's Law would be very needed!

You should write brief notes of how these techniques work (answers in the lecture notes 2b).

Question 5

The manufacturers of electric and hydrogen-powered vehicles have promoted their products as being environmental-friendly. However, this will not be true unless the electricity / fuels used to power these vehicles are produced from a renewable source. Why? Describe some ways of producing **clean electricity and hydrogen** (Please give the underlying science/chemistry of your choices). What makes your choices *clean*?

(5 marks)

Renewable electricity can be produced from turbines driven by wind and hydro (examples, dams) energy. Solar panels (photovoltaics) can be used.

The electricity that is produced can be stored (converted) in the form of fuels via electrochemical water splitting to produce H_2 gas. Details are in the notes.

Whatever the choice is, it is important that the energy input must be as green and sustainable as possible (this is most critical!).

CM5244 Topics in Environmental Chemistry (Semester 2, 2018/2019)

Test (15 April 2019)

Examiner: Assoc. Prof. Boon Siang Jason YEO

Time allowed: 45 min

Matric No: _____

INSTRUCTIONS

1. Please write your student number only. **Do not write your name**.
2. There are 5 questions. Answer all 5 of them.
3. Write your answers in the space provided below each question.

Question 1

(a) Write equations for the Chapman Cycle that describes the natural formation and destruction of stratospheric ozone.
(b) Suggest how HO• radicals could be formed in the stratosphere.
(c) Suggest how the presence of CH_4 in the stratosphere could also lead to the formation of additional HO• radicals.
(d) Write equations for stratospheric ozone destruction involving HO• catalyst.
(e) Do you think that the Chapman Cycle can be appropriately used to describe the formation of ozone found in a tropospheric photochemical smog? Rationalize your answer.

(10 marks)

(a) The Chapman cycle:

$O_2 + UV \rightarrow 2O$ *(step 1)*
$O + O_2 \rightarrow O_3 + heat$ *(step 2)*
$O_3 + (240\ nm < radiation < 310\ nm) \rightarrow O_2 + O$ *(step 3)*
$O_3 + O \rightarrow 2\ O_2$ *(step 4)*

(b) $O + H_2O \rightarrow HO$ (through the reaction of O atoms with water vapor)

(c) $O + CH_4 \rightarrow HO + CH_3$

(d) HO can destroy the ozone via mechanism I or II.
For example, Mechanism I:

$HO + O_3 \rightarrow HOO + O_2$
$HOO + O \rightarrow HO + O_2$

Overall reaction: $O_3 + O \rightarrow 2O_2$

(e) No. This is because most of the UV radiation that is required to photolyze the O_2 to O atoms (step 1 of Chapman cycle) would have been absorbed in the stratosphere.

Photolysis of NO_2 to give O, followed by O reacting with O_2, is the known way by which ozone can be produced in the troposphere.

$NO_2 \rightarrow NO + O$
$O + O_2 \rightarrow O_3$

Question 2

Both CFC-13 and HFC-23 have the same number of fluorine atoms. Therefore, they have the same tropospheric lifetimes.

Both CFC-13 and HFC-23 have the same number of fluorine atoms. Therefore, they have the same ozone depleting potentials when placed in the stratosphere.

Both CFC-13 and CO_2 have the same number of carbon atoms. Therefore, the per molecule ratio of radiative forcing of CFC-13 / CO_2 is expected to be 1.

Explain how the above three statements can be considered correct or are *fake news*!

(10 marks)

CFC-13 = $CClF_3$
HFC-23 = CHF_3

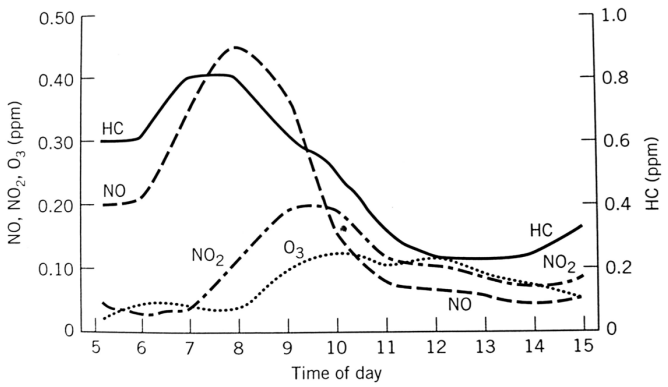
The tropospheric lifetime of HFC-23 is expected to be shorter than that of CFC-13, as the former has a C-H bond. Presence of a C-H bond allows HFC-13 to be attacked and destroyed by HO• radicals in the troposphere.

CFC-13 is expected to have a higher ozone depleting potential than HFC-23. This is because the former has a Cl atom which (once cleaved from the CFC-13) can destroy ozone. Fluorine atoms have no ozone depleting potentials.

CFC-13 have more vibrational modes than CO_2. Its vibrations associated with C-F and C-Cl bonds also lie in the atmospheric window (8000-13000 nm). This means that it can absorb the outgoing terrestrial radiation very efficiently. Hence, it is likely that the per molecule ratio of radiative forcing of CFC-13 / CO_2 is larger than 1.

Question 3

The concentrations of total hydrocarbons (HC), nitrogen oxides (NO and NO_2) and ozone (O_3) measured at a downtown location in an urban city called Selegna Sol for different hours of the day is shown below. Could you explain the origin of these molecules, and rationalize the time-dependent trends for their concentrations?



(10 marks)

The relevant chemical equations are shown in the lecture notes 3, Slides 9-17.

At the beginning of the day, NO and HC show a large decrease due to commuter traffic. After an hour or so, the NO has been converted to NO_2, and the photochemical reactions can now begin with the strong morning sun. The resultant O_3 peak about an hour later, and then diminish during the day (due to reduced traffic).

Note: Writing chemical equations in your answer would make it clearer.

Question 4

(a) Suggest strategy(ies) for reducing the concentration of NO_x in vehicle emissions. Include the relevant scientific principles of your strategy(ies).

(b) Describe an analytical method to *detect and quantify* the concentration of NO_x (both NO and NO_2) in air.

(10 marks)

Possible strategies include HDN catalysts for removing N from fossil fuels, selective catalytic reduction in diesel engines, and three-way catalytic convertor in petrol vehicles.

Chemiluminescence can be used to detect for NO. NO_2 can be converted to NO, before being detected by chemiluminescence. The use of Beer's law and standards has to be used to quantify the concentrations of the NO and NO_2. You should write brief notes of how the technique works (answer in the lecture notes 2b, 22-23).

Note: Writing chemical equations in your answer would make it clearer.

Question 5

'Our cars are solely powered by hydrogen (in hydrogen fuel cells) and would only produce H_2O emissions. Therefore, it is totally environmentally friendly'.

You found the above statement from a car manufacturer's website. It is clearly an over-simplification! To determine the environmental friendliness of the car, we also need to consider the entire process of its manufacture, method of producing the fuel, etc. Propose to the car manufacturer on how they could produce the hydrogen gas fuel sustainably and without any net positive carbon emissions. Include the chemical principles of your strategy.

(10 marks)

Water electrolysis can be used. This is performed in a water electrolyzer.

Note: a water electrolyzer is not the same as a fuel cell.

For the process to be truly sustainable, renewable electricity must be used to drive water electrolysis. This could be produced from turbines driven by wind, hydro (examples, dams) and geothermal sources. Electricity generated from solar panels (photovoltaics) can also be used.

CM5244 Topics in Environmental Chemistry (Semester 2, 2018/2019)

Test (15 April 2019)

Examiner: Assoc. Prof. Boon Siang Jason YEO

Time allowed: 45 min

Matric No: _____

INSTRUCTIONS

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Question 2

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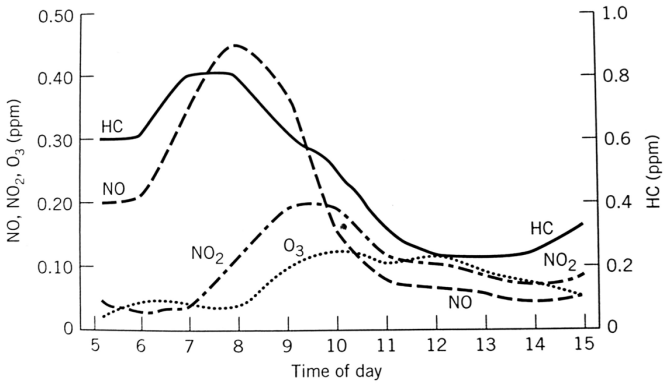

(10 marks)

Question 4

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(b) Describe an analytical method to *detect and quantify* the concentration of NO_x (both NO and NO_2) in air.

(10 marks)

Question 5

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You found the above statement from a car manufacturer's website. It is clearly an over-simplification! To determine the environmental friendliness of the car, we also need to consider the entire process of its manufacture, method of producing the fuel, etc. Propose to the car manufacturer on how they could produce the hydrogen gas fuel sustainably and without any net positive carbon emissions. Include the chemical principles of your strategy.

(10 marks)

CM5244 Topics in Environmental Chemistry (Semester 2, 2019/2020)

Test (17 April 2020)

Examiner: Assoc. Prof. Boon Siang Jason YEO

Time allowed: 1 hr, 15 min.

INSTRUCTIONS

1. Please write your student number only on your answer sheets. **Do not write your name**.

2. Page 2 of this test script contains the questions to answer in regard to the journal article by Lewis and Nocera. Please read these questions first, as it will help you to understand the article better.

3. The journal article of Lewis and Nocera starts from Page 3 of this test script.

4. Just like what you have done for the assignments, you can make use of your notes, textbooks, web resources and most importantly, your creativity, to answer the questions. However, please answer all the questions by yourself (NUS Code of Conduct).

5. You may type-out or hand-write your answers to the questions. Please present your answers in about 2 pages.

6. Please send me your answers through LumiNUS before 7:50 pm. You can upload a pdf, doc, photos, etc of your answers. (Upload to LumiNUS > Files > Submission > Test (17 April 2020)).

Questions

The review article by Lewis and Nocera in the Proceedings of the National Academy of Sciences (PNAS, 103, 15729 (**2006**)) is one of the most well-cited articles in the field of renewable energy and sustainability. It was published 14 years ago. Of course, this field has evolved rapidly since then, and one might expect some of the ideas in the article to be outdated or unpractical in Year 2020.

Here are two questions to answer. I will give you the liberty to answer these questions in whatever format you want. You can write one 'essay' that contains answers to these questions, or answer them separately. Your answers can also be in bullet point form. It is important to lay down the facts and arguments in a coherent and systematic fashion.

1. Read and summarize the key ideas of the article by Lewis and Nocera. Critique any ideas that you believe to be outdated or unpractical (taking your own home country, e.g., Singapore, China, India, etc. as the site for the implementation of these ideas). Please also give suggestions of how you might want to improve on the ideas of Lewis and Nocera.

2. The price of renewable electricity could fall to 1.5 US cents / KWh in the coming years! As a future leader in your country and company, what would you do with this cheap source of energy, in order to address what you think are the most pressing **sustainability** and **environmental problems** of the world? Please give the underlying science / chemistry of your ideas.

Powering the planet: Chemical challenges in solar energy utilization

Nathan S. Lewis*† and Daniel G. Nocera†‡

*Division of Chemistry and Chemical Engineering, California Institute of Technology, Pasadena, CA 91125; and
‡Department of Chemistry, Massachusetts Institute of Technology, Cambridge, MA 02139-4307

Edited by Edward I. Solomon, Stanford University, Stanford, CA, and approved August 11, 2006 (received for review May 25, 2006)

Global energy consumption is projected to increase, even in the face of substantial declines in energy intensity, at least 2-fold by midcentury relative to the present because of population and economic growth. This demand could be met, in principle, from fossil energy resources, particularly coal. However, the cumulative nature of CO_2 emissions in the atmosphere demands that holding atmospheric CO_2 levels to even twice their preanthropogenic values by midcentury will require invention, development, and deployment of schemes for carbon-neutral energy production on a scale commensurate with, or larger than, the entire present-day energy supply from all sources combined. Among renewable energy resources, solar energy is by far the largest exploitable resource, providing more energy in 1 hour to the earth than all of the energy consumed by humans in an entire year. In view of the intermittency of insolation, if solar energy is to be a major primary energy source, it must be stored and dispatched on demand to the end user. An especially attractive approach is to store solar-converted energy in the form of chemical bonds, i.e., in a photosynthetic process at a year-round average efficiency significantly higher than current plants or algae, to reduce land-area requirements. Scientific challenges involved with this process include schemes to capture and convert solar energy and then store the energy in the form of chemical bonds, producing oxygen from water and a reduced fuel such as hydrogen, methane, methanol, or other hydrocarbon species.

The supply of secure, clean, sustainable energy is arguably the most important scientific and technical challenge facing humanity in the 21st century. Energy security, national security, environmental security, and economic security can likely be met only through addressing the energy problem within the next 10–20 yr. Meeting global energy demand in a sustainable fashion will require not only increased energy efficiency and new methods of using existing carbon-based fuels but also a daunting amount of new carbon-neutral energy. The various factors that conspire to support the above far-reaching conclusions and the basic science needed for the development of a large-scale cost-effective carbon-neutral energy system are the focus of this paper.

The Global Energy Perspective

In 2001, worldwide primary energy consumption was 425×10^{18} J, which is an average energy consumption rate of 13.5 terawatt (TW) (1). Eight-six percent of this energy was obtained from fossil fuels, with roughly equal parts from oil, coal, and natural gas. Nuclear power accounted for ≈0.8 TW of primary (thermal) energy, and the remainder of the energy supply came mostly from unsustainable biomass, with a relatively small contribution from renewable sources (1).

Future energy demand is projected to increase considerably relative to that in 2001. The most widely used scenarios for future world energy consumption have been those developed by the Intergovernmental Panel on Climate Change, an organization jointly established by the World Meteorological Organization and the United Nations Environment Program (after Scenario B2 in ref. 2; $\dot{E}$ = (869 EJ/yr)·(10^6 TJ/EJ)/(60·60·24·365 s/yr) = 27.54 TW (TJ, terajoule; and EJ, exajoule). The scenario outlined in the last two columns of Table 1 is based on "moderate" assumptions and hence is reasonably viewed as neither overly conservative nor overly aggressive.

To better understand this scenario, the top half of Table 1 breaks down the rate of energy consumption, $\dot{E}$, into three fundamental factors (3):

$$\dot{E} = N\cdot(GDP/N)\cdot(\dot{E}/GDP), \quad [1]$$

where N is the global population, GDP/N is the globally averaged gross domestic product (GDP) per capita, and $\dot{E}$/GDP is the globally averaged energy intensity (i.e., the energy consumed per unit of GDP). The world population was ≈6.1 billion in 2001, and in the scenario represented in Table 1, the global population is projected to increase by 0.9% yr^{-1} to ≈9.4 billion by 2050. World per capita GDP was ≈$7,500 per capita in 2001. In the Table 1 scenario, GDP/N is projected to increase at the historical average rate of 1.4% yr^{-1} to ≈$15,000 per capita by 2050. No country has a policy against economic growth, so this increase in GDP/N seems quite reasonable and in fact may well be modest given the rapid economic growth being experienced by China and India at present. With no changes in the globally averaged energy intensity, the world energy consumption rate would grow, due to population growth and economic growth, by 2.3% yr^{-1}, from 13.5 TW in 2001 to ≈40.8 TW in 2050. However, the global average energy intensity has declined continuously over the past 100 yr, due to improvements in technology throughout the energy production, distribution, and end-use chain. In anticipation of continued improvements in technology, the global average energy intensity in the Table 1 scenario is projected to decrease at approximately the historical average rate of 0.8% yr^{-1}, from 0.29 W/($ yr^{-1}) in 2001 to 0.20 W/($ yr^{-1}) by 2050. This decrease offsets somewhat the projected increases in population and per capita GDP, so that the world energy consumption rate is instead projected to grow by 2.3% yr^{-1} − 0.8% yr^{-1} = 1.5% yr^{-1}, from 13.5 TW in 2001 to ≈27 TW by 2050. Hence, even factoring in a decrease in energy intensity, the world energy consumption rate is projected to double from 13.5 TW in 2001 to 27 TW by 2050 and to triple to 43 TW by 2100 (4).

The Global Energy Challenge Presented by Consumption of Fossil Fuels

Many sources indicate there are ample fossil energy reserves, in one form or another, to supply this energy at some reasonable cost. The World Energy Assessment Report estimates of the total reserves (i.e., 90% confidence that the

Author contributions: N.S.L. and D.G.N wrote the paper.

The authors declare no conflict of interest.

This article is a PNAS direct submission.

Abbreviations: TW, terawatt; GDP, gross domestic product; PV, photovoltaics; GtC, metric tons of carbon.

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Table 1. World energy statistics and projections

Quantity	Definition	Units	2001*	2050†	2100‡
N	Population	B persons	6.145	9.4	10.4
GDP	GDP§	T \$/yr	46	140¶	284‖
GDP/N	Per capita GDP	\$/(person-yr)	7,470	14,850	27,320
$\dot{E}$/GDP	Energy intensity	W/(\$/yr)	0.294	0.20	0.15
$\dot{E}$	Energy consumption rate	TW	13.5	27.6	43.0
C/E	Carbon intensity	KgC/(W·yr)	0.49	0.40	0.31
$\dot{C}$	Carbon emission rate	GtC/yr	6.57	11.0	13.3
$\dot{C}$	Equivalent CO_2 emission rate	$GtCO_2$/yr	24.07	40.3	48.8

*$\dot{E}$ = (403.9 Quads/yr)·(33.4 GWyr/Quad)·(10^{-3} TW/GW) = 13.5 TW; and $\dot{C}$ = (24.072 $GtCO_2$/yr)·(12/44 GtC/$GtCO_2$) = 6.565 GtC (adapted from ref. 1).

†$\dot{E}$ = (869 EJ/yr)·(10^6 TJ/EJ)/(60·60·24·365 s/yr) = 27.5 TW [adapted from ref. 2 (Scenario B2), pp. 48–55].

‡$\dot{E}$ = (1,357 EJ/yr)·(10^6 TJ/EJ)/(60·60·24·365 s/yr) = 43.0 TW [adapted from ref. 2 (Scenario B2), pp. 48–55].

§All in year 2000 U.S. dollars, using the inflation-adjusted conversions: $\$_{2000}$ = 1/0.81590 $\$_{1990}$ (adapted from ref. 1), and 'purchasing power parity' exchange rates.

¶In year 2000 U.S. dollars: (113.9 $T\$_{1990}$)·(1/0.81590 $\$_{2000}/\$_{1990}$) = 139.6 $T\$_{2000}$.

‖In year 2000 U.S. dollars: (231.8 $T\$_{1990}$)·(1/0.81590 $\$_{2000}/\$_{1990}$) = 284.1 $T\$_{2000}$.

reserves exist) and of the global resource base (5), including both conventional and unconventional sources, provide a benchmark for evaluating the total available global fossil energy base. Based on 1998 consumption rates, 40–80 yr of proven conventional and unconventional oil reserves exist globally, and 50–150 yr of oil are available if the estimated resource base is included. Sixty to 160 yr of reserves of natural gas are present, and between 207 and 590 yr of gas resources, not including the natural gas potentially available as methane clathrates in the continental shelves, are in the estimated resource base. Similarly, a 1,000- to 2000-yr supply of coal, shales, and tar sands is in the estimated resource base. Hence the estimated fossil energy resources could support a 25- to 30-TW energy consumption rate globally for at least several centuries.

Consumption of fossil energy at that rate, however, will produce a potentially significant global issue. Historically, the mean carbon intensity (kg of C emitted to the atmosphere as CO_2 per year per W of power produced from the fuel) of the global energy mix has been declining. In the past two centuries, the energy mix has shifted from being dominated by wood to coal to oil and now more to natural gas. This shift has produced a decrease in the average carbon intensity of the energy mix, because oil and gas have higher H/C ratios and hence upon combustion produce more water and less CO_2 per unit of heat released than does coal. If the carbon intensity were to remain at the year 2001 value (approximately equal parts coal, oil, and natural gas), the world carbon emission rate would grow due to the projected growth in the energy consumption from 6.6 billion metric tons of carbon (GtC) yr^{-1} in 2001 to 13.5 GtC yr^{-1} by 2050. The Intergovernmental Panel on Climate Change "business as usual" scenario of Table 1 projects, arguably optimistically, that the historical trend of mean carbon intensity decline with time will continue through 2050, producing an energy mix continually favoring cleaner-burning fuels from a carbon emissions viewpoint, until the average in 2050 is below that of the least carbon-intensive fossil energy source, natural gas. This decrease in carbon intensity would offset somewhat the increase in the rate of energy consumption. But even with this projected decrease in carbon intensity, the world carbon emissions rate in this scenario is projected to nearly double from 6.6 GtC yr^{-1} in 2001 to 11.0 GtC yr^{-1} by 2050 (2).

On the timescale of many centuries, CO_2 emissions are essentially cumulative in the atmosphere. The CO_2 equilibrates on an ≈10- to 30-yr timescale between the atmosphere and the near-surface layer of the oceans (6), which accounts for why only ≈50% of the anthropogenic CO_2 emissions remain in the atmosphere (the remainder partitioning into the biosphere and the oceans). Because there are no natural destruction mechanisms of CO_2 in the atmosphere, the long-term removal of atmospheric CO_2 must occur by convection. The relevant mixing time between the near-surface ocean layer and the deep oceans is between 400 and several thousand years (6, 7). Hence, in the absence of geoengineering or active intervention, whatever environmental effects might be caused by this atmospheric CO_2 accumulation over the next 40–50 yr will persist globally for the next 500–2,000 yr or more.

Although the precise future effects of such anthropogenic CO_2 emissions are still somewhat uncertain, the emission levels can certainly be viewed rigorously within a historical perspective. The data from the Vostok ice core indicate that the atmospheric CO_2 concentration has been between 210 and 300 ppm for the past 420,000 yr (8), and more recent studies of Dome Concordia ice cores have extended this time period to 650,000 yr (9). Over this same time period, the atmospheric CO_2 concentration has been highly correlated with, but is not necessarily the cause of, temperature swings that have repeatedly caused ice ages on the planet. The CO_2 concentrations in the past 50 yr have been rising because of anthropogenic CO_2 emissions from fossil fuel consumption, and they are now in excess of 380 ppm. Without intervention, even the Table 1 scenario produces, within the 21st century, atmospheric CO_2 concentrations that are more than double the preanthropogenic values (4, 6). The exact levels vary depending on the assumed composition of energy sources, the efficiency of energy production and consumption, the global economy, and different intervention scenarios to control CO_2 levels. Modestly stringent interventions are based on stabilizing atmospheric CO_2 in the 550- to 650-ppm range, with substantially higher values projected (>750 ppm) if the Table 1 scenario is followed. Climate models predict a variety of different global responses to levels of CO_2 at or in excess of 550 ppm in the atmosphere. In some models, moderate changes are predicted, whereas in others, relatively serious sea level rises, changes in the hydrological cycle, and other effects are predicted (10). Tipping points involving positive feedback, such as the accelerated loss of permafrost, which could release further CO_2 which then could accelerate still further permafrost loss, are of substantive concern. What can be said with certainty is that the atmospheric CO_2 concentrations are being increased and without severe intervention will con-

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tinue to increase, because of anthropogenic sources, to levels that have not been present on the planet in at least the past 650,000 yr and probably in the past 20 million yr.

Carbon-Neutral Fuel Sources and the Solar Opportunity

To meet the (arguably optimistic) Intergovernmental Panel on Climate Change projection in the Table 1 scenario for the average carbon intensity in 2050, the projected carbon intensity in 2050 is $\approx$0.45 kg of C yr^{-1} W^{-1}, which is lower than that of any of the fossil fuels. The only way one can reach this value of the mean carbon intensity is through a significant contribution of carbon-free power to the total energy mix. This conclusion holds for an economy entirely based on natural gas; to the extent that the mix of consumed fossil fuels is not 100% natural gas but is roughly also equal parts oil and coal, even more carbon-free energy is required to maintain the average of the energy mix at the 0.45 kg of C yr^{-1} W^{-1} value. In fact, the amount of carbon-free power required in 2050 to meet these carbon intensity targets is >10 TW and is much greater than 10 TW if emissions are to be lowered such that CO_2 can be stabilized at 550 ppm. Even more carbon-free power will be required later in the 21st century if CO_2 levels are to be kept below 550 ppm or if a lower atmospheric CO_2 target level is desired. By almost any reasonable estimate, stabilization of atmospheric CO_2 levels at 550 ppm or lower will require as much carbon-neutral power by approximately the year 2050 as the amount of power produced at present from all energy sources combined (4). Furthermore, because CO_2 emissions are cumulative on a century-level timescale, even higher levels of carbon-neutral power are required by 2050 if their introduction does not start immediately with a constant rampup but instead are delayed by 20 yr for their commissioning while awaiting technology development and/or policy and socioeconomic interventions.

Three general routes are available to produce such large amounts of carbon-neutral power.

Nuclear fission is one method, but it would require widespread implementation of breeder reactors (11). Estimated terrestrial U resources are sufficient to produce $\approx$100 TW-yr of electricity using conventional once-through U reactor technology. Hence, if 10 TW of power were obtained from conventional nuclear fission, the terrestrial U resource base would be exhausted at that level in less than a decade (in fact, it would be exhausted after the first 30 yr of reactor construction because of the fuel consumed during the rampup phase). Moreover, construction of nuclear power plants would need to proceed at a very rapid rate by historical standards (one 1-GW_e (gigawatt-electric) power plant every 1.6 days for the next 45 yr). The international tokamak (magnetic confinement fusion) experiment (ITER) is now scheduled to demonstrate an energy breakeven point in 35 yr for a few minutes of operational time. Although fusion might possibly provide significant commercial energy late in the 21st century, the ITER time line is much too far in the future to provide a credible option to make a significant contribution to the amount of cost-effective carbon-neutral energy production needed to meet any reasonable atmospheric CO_2 concentration target in the next 40–50 yr.

Carbon capture and storage comprise a second general approach (12). In this approach, the carbon dioxide is dissolved in the underground aquifers. To be a viable option technically, the CO_2 must not leak at a globally averaged rate of 1% for a timescale of centuries. Otherwise, the emitted flux will be greater than or equal to that intended to be mitigated initially. Experiments at scale are needed, along with extensive modeling, simulation, monitoring, and validation, to ascertain with >99% confidence that the leak rate will be acceptably low for a 500- to 1,000-yr period. Furthermore, each reservoir is different geologically, so proof that sequestration works technically at one reservoir is not general proof that the process will work at the required level globally. The global reservoir capacity has been estimated to be equivalent to $\approx$100–150 yr of carbon emissions. Hence, sequestration could buy time if it works technically and is so validated within the next 10–20 yr. An additional condition is that the energy distribution and end-use chain must be transformed to handle massive quantities of carbon-free fuels (hydrogen) or electricity on the needed timescale to mitigate carbon emissions.

The third general approach is to use renewable energy. Of the various renewable energy sources, by far the largest resource is provided by the sun. More energy from sunlight strikes the earth in 1 hr (4.3×10^{20} J) than all of the energy currently consumed on the planet in 1 yr (4.1×10^{20} J in 2001) (5). Yet, in 2001, only <0.1% of electricity and <1.5% of fuels (mostly from biomass) were provided by a solar source (1). Against the backdrop of the daunting carbon-neutral energy needs of our global future, the large gap between our present use of solar energy and its enormous undeveloped potential defines a compelling imperative for science and technology in the 21st century.

Solar Energy Utilization

Solar energy utilization requires solar (*i*) capture and conversion and (*ii*) storage. Solar capture and conversion may be accomplished by photovoltaics (PVs). The challenge here is to dramatically reduce the cost per W of delivered solar electricity. Compared with fossil energy, solar energy is diffuse, and hence materials costs must be very inexpensive to make a solar-based process economical. Knowing the insolation striking an area of the earth for a 30-yr period, it is relatively simple to calculate the sale price of the converted energy that is needed to pay back at least the initial cost that is required to cover that area with the solar energy conversion system. At 10% efficiency, and a cost of \$300 m^{-2}, both typical of current Si-based solar electricity modules, along with a balance of systems cost of \$3 W_p^{-1} (peak W), an electricity price of \$0.35 $[\text{kW-hr}]^{-1}$ is required to cover the initial system costs (13). By comparison, fossil-derived electricity (high-value energy) now costs approximately \$0.02–0.05 $[\text{kW-hr}]^{-1}$, and that cost includes storage and distribution costs. To reach a cost point near that of fossil-derived energy will thus require improvements in efficiency but additionally will require large decreases in cost, into a range below \$100 m^{-2}. For comparison, the cost of paint is about \$1 m^{-2}, so the solar energy conversion system can cost $\approx$10 times more than the cost of paint, but not much more if it is to provide cost-effective primary energy.

In the absence of cost-effective storage, solar electricity can never be a primary energy source for society, because of the diurnal variation in local insolation. In principle, storage of electricity could be obtained using batteries, but at present no battery is inexpensive enough, when amortized over the 30-yr lifetime of a solar device, to satisfy the needed cost per W targets for the whole system. A second method is to store the electrical energy mechanically. For instance, electricity could be used to drive turbines to pump water uphill. This approach is relatively inexpensive for storing large amounts of energy at modest charge and discharge rates, but is not well matched to being charged and discharged every 24 h to compensate for the diurnal cycle. For example, buffering the day/night cycle in the U.S. energy demand by this approach would require a pumping capacity equivalent of >5,000 Hoover Dams, filling and emptying reservoirs every day and every night. Currently, the cheapest method of solar

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energy capture, conversion, and storage is solar thermal technology, which can cost as little as $0.10–0.15 per kW-hr for electricity production. Advances in this potentially very important approach to solar energy utilization will require new materials for the focusing and thermal capture of the energy in sunlight, as well as new thermochemical cycles for producing useful fuel from the captured solar energy. The possibility of integrated capture, conversion, and storage functions makes solar thermal technology an option that should be vigorously pursued to exploit the large untapped solar energy resource for carbon-neutral energy production

A third method of storage is to borrow the design of nature, in which chemical bonds are broken and formed to produce solar fuels in an artificial photosynthesis process. Photosynthesis itself is relatively inefficient, when measured on a yearly average basis per unit area of insolation. For example, switchgrass, one of the fastest-growing crops, yields energy stored in biomass at a yearly averaged rate of <1 W/m^2 (5). Because the averaged insolation at a typical midlatitude is 200–300 W/m^2 (5), the yearly averaged energy conversion and storage efficiency of the most rapidly growing large area crops is currently <0.5%. Even if this efficiency could be reached with no energy inputs into the farm or any energy losses due to outputs from the utilization of the biomass, growth of energy crops on all of the naturally irrigated cultivatable land on earth that is not currently used for food crops would yield perhaps 5–10 TW of total power. Whereas biofuels derived from existing plants could provide a potentially significant contribution to liquid fuels for transportation uses (if cellulosic conversion technology can be successfully developed and deployed economically) increased energy conversion and storage efficiency are highly desirable to remove land area as a serious constraint on the amount of energy that can be obtained from the sun and stored in chemical bonds. One approach is to develop an artificial photosynthetic process with an average efficiency significantly higher than plants or algae.

The primary steps of photosynthesis involve the conversion of sunlight into a "wireless current." In all cases, to form a useful fuel, O_2 must be evolved, so it can be released into our oxygen-containing atmosphere and used elsewhere as an oxidation reagent for fuel consumption. The reduced fuel could be either hydrogen from water reduction, or it could be an organic species, such as methanol or methane, that is derived from the fixation of atmospheric CO_2. Recombination of the reduced fuel with released O_2 would then regenerate the original species, closing the cycle in a carbon-neutral fashion.

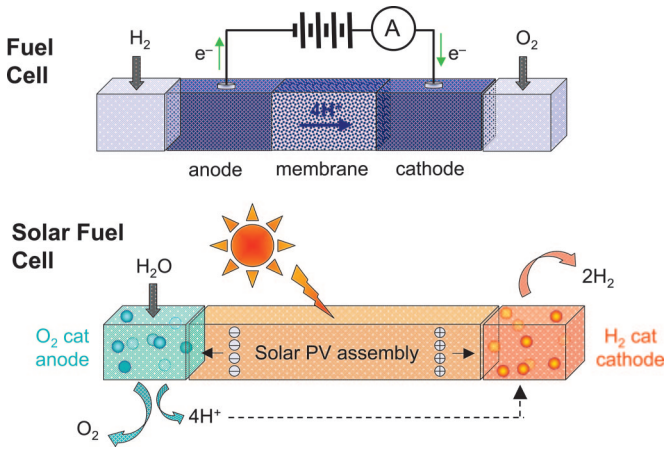

Fig. 1. H_2 and O_2 are combined in a fuel cell to generate a flow of electrons and protons across a membrane, producing electrical energy. The solar fuel cell uses light to run the electron and proton flow in reverse. Coupling the electrons and protons to catalysts breaks the bonds of water and makes the bonds H_2 and O_2 to effect solar fuel production.

In natural photosynthesis, the anodic charge of the wireless current is used at the oxygen-evolving complex to oxidize water to oxygen, with the concomitant release of four protons. The cathodic charge of the wireless current is captured by Photosystem I to reduce the protons to "hydrogen," with the reduced hydrogen equivalents stored through the conversion of NADP to NADPH. Thus, the overall primary events of photosynthesis store sunlight by the rearrangement of the chemical bonds of water, to form oxygen and Nature's form of hydrogen.

An artificial photosynthetic system could be realized by spatially separating solid-state or molecular water reduction and oxidation catalysts and connecting them to a light collection and charge separation system. In one such construct, the spatially separated electron–hole pairs provided by a photovoltaic assembly based on a solid-state junction, on either the macroscale or the nanoscale, are captured by the catalysts, and the energy is stored in the bond rearrangement of water to H_2 and O_2. Other concepts involve more intimate integration of the charge separation and chemical bond-forming functions, to avoid costs and system constraints associated with electrical contacts, wires, inverters, etc., involved with converting 1-eV photons into 1-eV chemical bonds through electricity as a discrete intermediary. One approach to this type of system is depicted in Fig. 1, in which the tightly integrated system is modeled after natural photosynthesis and serves as a model for the artificial photosynthetic systems that are discussed below.

The Basic Science Needs for PVs

One of the key issues in solar capture and conversion is how to separate charge efficiently over macroscopic distances without using expensive, highly pure, semiconductor materials. This effort requires the development of new chemical and materials methods to make polycrystalline and nanocrystalline semiconductors perform as if they were expensive single crystals. Numerous research approaches are being pursued (13). Materials consisting of a network of interpenetrating regions can facilitate effective charge separation and collection, thus relaxing the usual constraint in which the photogenerated carriers must exist long enough to traverse the entire distance of the cell. Present photon conversion devices based on a single-bandgap absorber, including semiconductor PV, have a theoretical thermodynamic conversion efficiency of 32% in unconcentrated sunlight. However, the conversion efficiency can be increased, in principle, to 45–65% if carrier thermalization can be prevented (by overcoming the so-called Shockley–Queisser limit). Multiple-bandgap absorbers in a cascaded junction configuration can result in high photoconversion efficiencies, particularly when cells are designed to sustain the operating conditions (e.g., elevated temperatures) associated with highly concentrated sunlight. It is expected that mature high-concentration PV systems can provide 10–20% more energy than standard PV systems with the same installed power rating.

In addition to making evolutionary changes to existing PV technologies,

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new materials for next-generation PVs are needed. Building upon the recent success in developing efficient molecular organic PVs and the recent advances in the controlled assembly of hybrid organic/inorganic nanostructures, organic and hybrid PV cells could possibly exceed 10% energy conversion efficiency, while offering a potentially inexpensive manufacturing paradigm (e.g., casting from emulsions, printing, and use of flexible substrates for production of large-area thin-film cells; ref. 14). To guide the PV nanostructure assembly, biologically derived and/or genetically engineered systems might be used to control the crystal structure, phase, orientation, and nanostructural regularity of inorganic materials. Genetically modified photosynthetic complexes from plants and bacteria can also convert incident light into photocurrent. Although the present energy conversion efficiencies of such systems are low, the projected maximum could be possibly as high as 10%. Finally, the Shockley–Queisser limit may be overcome by using multilayer junctions of semiconductor quantum dots, quantum wells and related nanostructures, and new inorganic materials and photoassemblies. Such materials could channel the excess energy of electron/hole pairs into photovoltages and photocurrents, with the design guided by a refined detailed understanding of photon absorption, charge creation, and charge separation processes.

The Basic Science Needs for Solar Fuels

As described above, an important storage approach involves conversion of the energy captured in the charge-separated states of a solar capture and conversion system into chemical bonds. Water splitting is an example of a more general conversion to a solar fuel cycle that involves evolution of oxygen as one component and formation of a reduced fuel as the other. Unexplored basic science issues are immediately confronted when the problem is posed in the simplest chemistry framework (see Scheme 1).

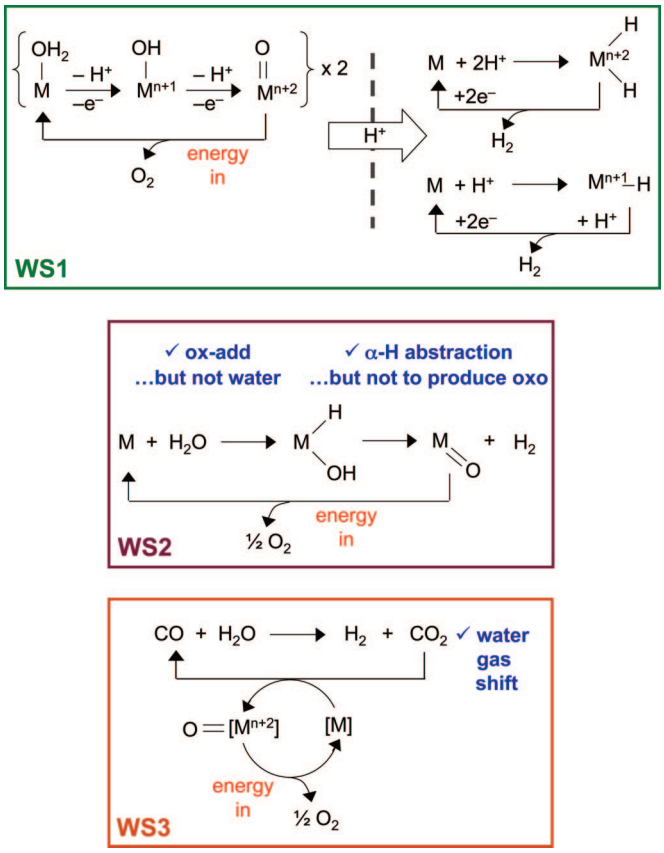
The overall transformation is a multielectron process promoted by photocatalyst and light. Elucidation of the fundamental principles of single electron-transfer reactions represented

$$2H_2O \xrightarrow[PC]{h\nu} 2H_2 + O_2$$

$$4H^+ + 4e^- \longrightarrow 2H_2$$

$$2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$$

Scheme 1.



Scheme 2.

such an important milestone in chemistry that two Nobel Prizes were awarded for such work (15, 16). Although dramatic advances have occurred in our understanding of single electron-transfer reactions, especially those in biology (17), a similar level of understanding of multielectron redox reactions has yet to be realized. Moreover, to ensure charge neutrality in the system, proton transfer must accompany electron transfer (i.e., proton-coupled electron transfer; ref. 18); hence, electron and proton inventories both need to be managed (19). Water splitting additionally presents sizable thermodynamic and kinetics barriers to making and breaking the bonds required to facilitate the desired chemical reactions. This is especially pertinent to the water-splitting problem, because the byproduct of water activation at the catalyst, whether molecular or solid, will invariably yield species that have strong metal–oxygen bonds. To close a catalytic cycle, these stable bonds need to be activated by the captured solar energy either directly or indirectly. More generally, the activation of all small molecules of consequence to carbon-neutral solar energy storage, including CO_2, O_2, and H_2O, share the reaction commonalities of bond-making and -breaking processes that require multielectron transfers coupled to proton transfer.

The Reaction Chemistry of Solar Energy Storage in Chemical Bonds

Perhaps the most straightforward water-splitting scheme is to have catalysts act directly on water, as exemplified by the two half-reactions denoted as WS1 (WS1, water-splitting strategy 1) in Scheme 2. The spatial separation of the catalysts requires that the charge-separation function be imbedded in some type of membrane, so that the protons generated on the anodic side of the cell are transported to the cathodic side of the cell for reduction. In effect, the system must be run in the opposite direction of a fuel cell, with sunlight providing the thermodynamic impetus to drive the cell in the desired fuel-forming direction.

The preparation of hydrogen-producing catalysts constitutes an intense area of study. Fe-only hydrogenases, comprised of small dithiolate-bridged bimetallic iron cofactors coordinated by CO and cyanide ligands, provide a benchmark for the efficient evolution of H_2 in molecular systems (20, 21). Structural, and in some cases, functional, analogues

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of such enzymatic active sites have been prepared (22–27); however, none of these Fe-only hydrogenase biomimics yet produce H_2 efficiently at low overpotential. Synthetic catalysts compare favorably to, and in some cases exceed, the efficiency of the biomimetic models. In the presence of sacrificial chemical reductants, mononuclear and binuclear metal complexes of Co, Ni, and Rh are known to effect catalytic hydrogen evolution electrochemically or photochemically (28–36). Intimate mechanistic details, however, are known in only a few cases (37), and the different possibilities, such as protonation of a hydride vs. uni- or bimolecular reductive elimination (right side, WS1, Scheme 2), in general have not yet been unraveled.

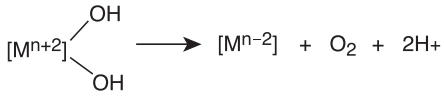
Other water-splitting cycles can also be developed. The water-splitting schemes WS2 and WS3 presented in Scheme 2 use basic reaction types that are common to organometallic catalysis. However, for the water-splitting problem, O, as opposed to C or N, needs to be managed. Every reaction, however, does have a precedent for carbon or nitrogen. In WS2 in Scheme 2, oxidative addition across X—H (X = C, N) bonds is a basic reaction of organometallic chemistry but is not yet well established for water (38–43). If this reaction can be achieved cleanly, hydrogen may be generated by α-H abstraction, which is a common reaction in organometallic chemistry and is used to generate metal–ligand multiple bonds. For instance, the α-H abstraction of metal alkylidenes produces alkylidynes (44). But α-H abstraction to produce metal-oxo species, and H_2 is uncommon for well defined hydroxo–hydrido complexes. In the case of WS3, the water–gas shift reaction produces H_2 from H_2O using CO as the reductant. An intense research effort, beginning in the 1970s and ending in the 1980s, provided the basic science for the development of catalysts to effect the water–gas shift reaction (45). However, the reaction must be closed by the conversion of CO_2 to CO. On this front, little is known. Some inroads to CO_2 reduction have been made on photo– (46, 47) and electro– (48–50) catalytic fronts, but generally the precise path to CO_2 reduction is ill-defined, making it difficult to improve these systems by design. A recent report of CO_2 reduction by a well defined homogeneous metal complex operating at high turnover number and frequency (51) is a harbinger of the promise that basic science holds for the design of efficient CO_2 reduction catalysts.

As in WS1, WS2 and WS3 cycles are closed by oxygen production, providing a further imperative for the development of reactions of the type described by Schemes 3–5. However, very few catalysts are known to oxidize water near the thermodynamic potential. Again, the most notable system is in biology, specifically involving the oxygen-evolving complex (OEC) of Photosystem II. The OEC comprises a cluster of four Mn centers and a Ca center (52–54), but no functional or structural models of the catalytically active site are yet available (55). At present, the ruthenium dimer $[(bpy)_2(OH_2)Ru^{III}ORu^{III}(OH_2)(bpy)_2]^{4+}$ (bpy = 2,2′-bipyridine) (56) and its relatives (57–59) represent the only unequivocally established molecular electrocatalysts for generating O_2 from H_2O. However, at present, this reaction proceeds at a high overpotential and with modest turnover numbers.

Scheme 3.

The success of WS1, WS2, and WS3 and other yet-undefined water-splitting schemes is predicated on systems that promote the conversion of oxygen from metal oxos. Many mechanistic possibilities for this conversion await discovery. They include the following.

(*i*) Nucleophilic Attack of Hydroxide on High-Valent Metal Oxos (Scheme 3). This basic reaction type is the foundation on which oxidation catalysts have been developed in the disciplines of organometallic and organic chemistry (60). Here the olefin bond attacks a metal oxo species to form two carbon–oxygen bonds. The replacement of the two-electron bond of the olefin by the lone pair of hydroxide would lead to the oxygen–oxygen bond-forming reaction that is critical for water oxidation. The substitution, however, is not trivial. OH^- is thermodynamically more difficult to oxidize than are olefins. Also, the overall reaction to produce oxygen involves a four-electron change at the metal, so there may be benefits to examining reductive elimination from more than one metal center, in which the multielectron equivalency can be shared by metal centers working in concert.

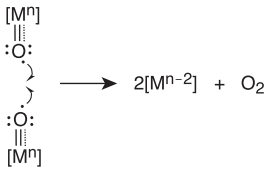


Scheme 5.

(*ii*) Radical Coupling of Two Oxos (Scheme 4). As shown, the oxygen radical may be delivered directly from a high-valent metal. Alternatively, the oxo species could be delivered from a multiply bonded metal–ligand species (61). This latter approach represents a paradigm shift in oxygen chemistry, because the strong metal–oxo double and triple bonds may be avoided, potentially lowering the activation barrier for oxygen atom delivery from a reactive multiply bonded metal–ligand center.

(*iii*) Reductive Elimination of bis Hydroxos or bis μ-Oxos (Scheme 5). These unknown reaction types encompass a four-electron change to make oxygen. A shared electron equivalency among a multimetallic center may expedite the reaction, such as that shown for Scheme 6.

In Scheme 2, the WS cycles are completed by the same parent metal complex. This does not have to be the case. As has recently been demonstrated, metal complexes working in tandem can promote reactions of energy consequence (62). Accordingly, the water-splitting schemes may be accomplished by two different metal complexes working in concert. Regardless of the precise details of the reaction design, oxygen production invariably will be an energetically demanding process that must be coupled to a charge-separated state to capture, convert, and store solar energy in the form of chemical bonds. By use of a photovoltaic assembly to accomplish solar-driven charge separation, the constraints on the catalyst design are relaxed solely to provide storage. However, in bringing catalysts to a charge-separating assembly, the reaction chemistry will be performed in a heterogeneous and/or interfacial environment. Accordingly, the need to acquire a molecular-level understanding of reactions at the surfaces of solids represents another scientific challenge confronting the effective utilization of solar energy. Finally, inasmuch as the aforementioned reactions and schemes are all enacted at a metal-based platform, the role of inorganic chemistry, whether at a



Scheme 4.

Scheme 6.

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molecule or a surface, will be pivotal to the development of the aforementioned water-splitting cycles. Ingenious approaches to water splitting may be possible using organic catalysts and biocatalysts as well, although the ability to operate these reactions at low overpotential will represent a significant challenge.

Conclusions

The sun has a unique role in sustainable energy production, in that it is the undisputed champion of energy; the resource base presented by terrestrial insolation far exceeds that of all other renewable energy sources combined. The solar energy resource additionally far exceeds what can possibly be envisioned as a level of human consumption necessary to support even the most technologically advanced society. However, to be a material contribution to primary energy supply, solar energy must be captured, converted, and stored to overcome the diurnal cycle and the intermittency of the terrestrial solar resource. Arguably the most attractive method for this energy conversion and storage is in the form of chemical bonds, by production of cheap solar fuels. Significant advances in basic science, however, are needed for this technology to attain its full potential. Chemistry will assume a special role in this endeavor, because new materials must be created for solar capture and conversion, and because new catalysts are needed for the desired chemical bond conversions. Here we present a blueprint for a reaction chemistry, when interfaced to a charge-separation structure, that permits artificial photosynthesis to be envisioned. The progress of scientists in chemistry, biology, engineering, materials science, and physics in addressing the basic science challenges involved with realizing this artificial photosynthesis will be critical to enable humans to use the sun sustainably as their primary energy source.

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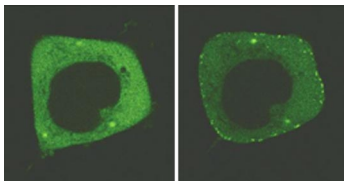
Corrections

IN THIS ISSUE, MEDICAL SCIENCES. For the "In This Issue" summary entitled "Carvedilol sidesteps G proteins," appearing in issue 42, October 16, 2007, of *Proc Natl Acad Sci USA* (104:16392), the figure caption appeared incorrectly. The online version has been corrected. The figure and its corrected caption appear below.



Carvedilol recruits β-arrestin to the β2-adrenergic receptor. The β-arrestin2-GFP is shown in green.

www.pnas.org/cgi/doi/10.1073/pnas.0710562104

PERSPECTIVE. For the article "Powering the planet: Chemical challenges in solar energy utilization," by Nathan S. Lewis and Daniel G. Nocera, which appeared in issue 43, October 24, 2006, of *Proc Natl Acad Sci USA* (103:15729–15735; first published October 16, 2006; 10.1073/pnas.0603395103), the authors note that in Fig. 1, the charges shown in the solar fuel cell are on the wrong sides of the cell. The holes should be at the anode, and the electrons should be at the cathode. This error does not affect the conclusions of the article. The corrected figure and its legend appear below.

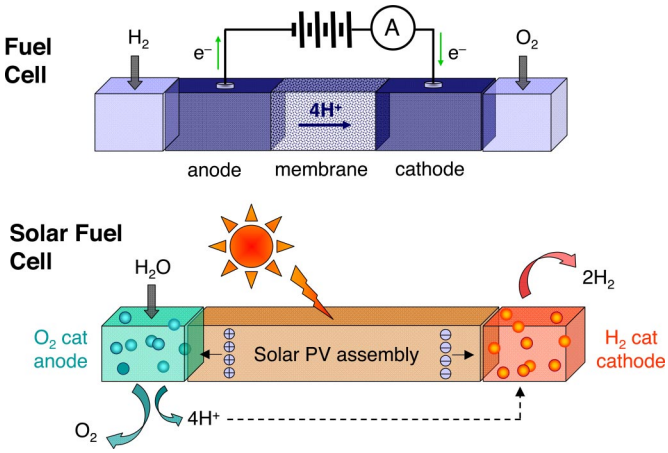


Fig. 1. H_2 and O_2 are combined in a fuel cell to generate a flow of electrons and protons across a membrane, producing electrical energy. The solar fuelcell uses light to run the electron and proton flow in reverse. Coupling the electrons and protons to catalysts breaks the bonds of water and makes the bonds H_2 and O_2 to effect solar fuel production.

www.pnas.org/cgi/doi/10.1073/pnas.0710683104

BIOPHYSICS. For the article "Drift and breakup of spiral waves in reaction–diffusion–mechanics systems," by A. V. Panfilov, R. H. Keldermann, and M. P. Nash, which appeared in issue 19, May 8, 2007, of *Proc Natl Acad Sci USA* (104:7922–7926; first published April 27, 2007; 10.1073/pnas.0701895104), the authors note that on page 7922, right column, the first sentence in *Mathematical Model*, "Our RDM model is based on a three-variable Fenton–Karma RD model for cardiac excitation (15), coupled with the soft-tissue mechanics equations described in refs. 12 and 16 . . . , where $\Theta(x)$ is the standard Heaviside step function: $\Theta(x) = 1$ for $x \geq 0$ and $\Theta(x) = 0$ for $x < 0$," should instead read: "Our RDM model consists of RD equations developed by F. H. Fenton (personal communication) and is based on a three-variable Fenton–Karma RD model for cardiac excitation (15), coupled with the soft-tissue mechanics equations described in refs. 12 and 16 . . . , where $\Theta(x)$ is the standard Heaviside step function: $\Theta(x) = 1$ for $x \geq 0$ and $\Theta(x) = 0$ for $x < 0$." Additionally, on page 7923, left column, beginning on line 10 of the text, the formula for I_{si} is incorrect in part. The portion of the formula appearing as "$(0.46 + 0.085 \cdot \tan h[k(u - 0.5)])$" should instead appear as: "$(0.23 + 0.085\tanh[10(u - 0.65)])$." Thus, the corrected formula should read $I_{si}(u, w) = \Theta(u - 0.2)uw(0.23 + 0.085\tanh[10(u - 0.65)])$. Finally, on page 7926, in the first sentence of the Acknowledgments, the authors would like to more specifically acknowledge the assistance of Dr. Fenton. Therefore, "We thank Dr. F. Fenton, Prof. P. J. Hunter, and Dr. P. Kohl for valuable discussions" should instead read: "We are grateful to Dr. F. H. Fenton, who kindly provided equations used in the construction of our RDM model, and to Prof. P. J. Hunter and Dr. P. Kohl for valuable discussions." These errors do not affect the conclusions of the article.

www.pnas.org/cgi/doi/10.1073/pnas.0710559104

Dear all,
Regarding your test's performance……it is, **so far, so good**!
All the best for the rest of your tests and exams for other modules!
Best,
Jason

Some guidelines to the answers for the test. I give broad flexibility to how you want to write your answers. But the ideas and facts should be coherently laid down.

I have put down a few points below. Maybe your answers do not contain some of them. But this, by no means, would imply that your answers are wrong. It could just mean that you are thinking more than me.

Answer to Q1:

#1: solar electricity

Electricity produced using photovoltaics (solar panels) have dramatically fallen, and have reached grid parity. Hence, it is no longer as Lewis and Nocera said '**a *challenge..to dramatically reduce the cost per W of delivered solar electricity***'.

It is notable that in the case of Singapore, due to our geographical size, it is not possible to be fully powered by solar panels. You can do back-of-envelope calculations to figure out how much solar electricity can be produced each year in Singapore, and relating this number to the amount of electricity Singapore consumes annually (**49.6 TWh in 2018, see slide 34 of your lecture 5 notes.)**. Of course, please do realistic calculations, and not assume that the whole of Singapore is covered by a big solar panel!

#2: carbon capture and storage

Carbon capture and storage will never be a viable lasting solution, as it only delays the fact that at some point in time, we have to deal with the excess carbon dioxide that we are producing. It is also not practical in the case of Singapore, due to our urban buildup. **Carbon capture and conversion** will be a more practical strategy.

#3: Artificial photosynthesis

A subtle point that is not mentioned in the paper (and also in our lectures) is the problem of separation / purifications of products. For example, if we try to produce ethanol using CO_2 reduction, the ethanol would be in mixture of water and other CO_2 reduction products. Energy would be required to separate and extract the ethanol.

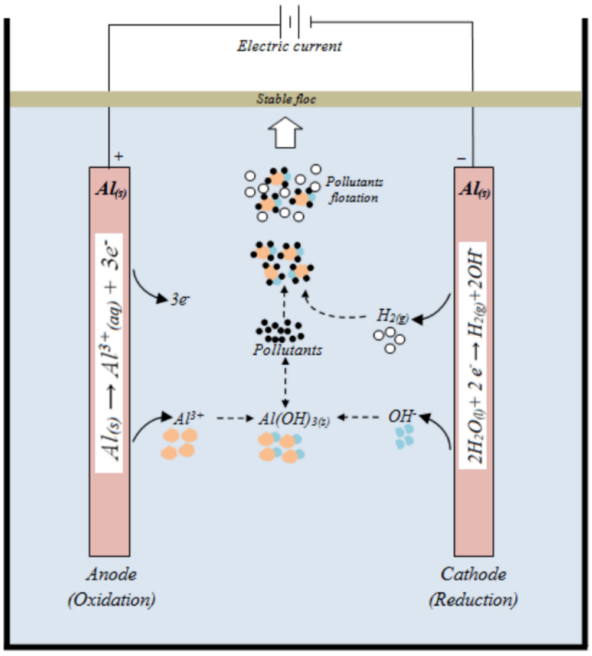
Answer to Q2:

The 'given' here is that renewable electricity would be cheap, and you are asked to make good use of this electricity to address what you think are the most pressing **sustainability** and

environmental problems of the world. This question is not asking for ways to produce more renewable electricity, but rather, how to use it.

You have flexibility here to identify on what you think the problems are, and the solutions. It will be nice to describe the science behind these solutions. In general, many of these problems could be solved by replacing the energy source, i.e., fossil fuels, with renewable electricity.

- For example, water purification using electrochemical methods. These could be electrodialysis, electrocoagulation, etc. Some of these technologies are listed in the paper - Environ. Sci.: Water Res. Technol., 2, 800 (**2016**). For the science behind these solutions, you could draw a picture to describe it. For example, in the case of electrocoagulation, you could draw its setup:



- The cheap electricity can also be used for transportation, specifically, to charge the batteries used for electric vehicles.
- The cheap electricity can also be used to convert carbon dioxide to chemicals and fuels, and reduce water to hydrogen. You can draw up schemes to illustrate the principles behind these setups.

CM5244 Advanced Environmental Chemistry (Semester 2, 2022/2023)

Test (23rd March 2023)

Examiner: Assoc. Prof. Boon Siang Jason YEO

Time allowed: 1 h, 15 min

Student No: _____

INSTRUCTIONS

1. Please write your student number only. **Do not write your name**.
2. Answer all questions.
3. Write your answers in the space provided below each question.
4. Open book format: You may consult notes/ textbooks (only hard copies). Electronic devices such as laptops / mobile phones / ipads are not allowed.
5. You may use a calculator.
6. There are 10 printed pages in this script.

Question 1

Explain how each of the four statements below can be considered correct or incorrect.

(a) 'The use of the Chapman Cycle alone to model the concentration of stratospheric ozone tends to lead to an under-estimation of the actual (stratospheric ozone) concentration.'

(2.5 marks)

Incorrect. The Chapman cycle does not take into account that there are other mechanisms for the destruction of ozone, such as those that involve Cl radicals from CFC. Hence, the use of Chapman cycle alone will lead to an over-estimation of the actual stratospheric ozone concentration (Lecture 1 notes, Slide 37).

(b) 'The Chapman Cycle can be appropriately used to model the concentration of tropospheric ozone, although it tends to lead to an over-estimation of the actual (tropospheric ozone) concentration.'

(2.5 marks)

Incorrect. The Chapman cycle cannot be appropriately used to model the tropospheric ozone concentration, as there is not enough UV-B reaching ground level. Tropospheric ozone concentration is better modelled by examining the concentration of NO_2, intensity of visible light, concentration of OH radicals, etc. (Lecture 1 notes, Slide 34; this was mentioned during the lecture) .

(c) 'Increased presence of hydrocarbons in the stratosphere is a leading cause of stratospheric ozone destruction!'

(2.5 marks)

Incorrect. Hydrocarbons are not likely to be leading cause of stratospheric ozone destruction, as these tend to be easily photolyzed and oxidized to CO_2 in the stratosphere.
(Lecture 1 notes, Slide 41)

(d) 'Chlorine from swimming pools is a source of chlorine radicals that destroy stratospheric ozone!'

(2.5 marks)

Incorrect. Cl_2 from the swimming pool will not be able to reach the stratosphere, as it dissolves easily in moisture, i.e., it has a tropospheric sink.

Question 2

(a) Is the following statement correct or incorrect? Rationalize your answer.

'On a per molecule basis, CFC-113 is likely to cause a higher positive radiative forcing compared to CO_2, because CFC-113 has one more carbon atom than CO_2.'

(3 marks)

Using the rule of 90, CFC-113 has 2 C atoms, 0 H atoms and 3 F atoms. Hence, its molecular formula is $C_2F_3Cl_3$. It has 3(8)—6=18 vibrational modes. It is also less symmetrical compared to CO_2. From this, we can already conclude that CFC-113 has more infrared active vibrations, and that it absorbs more IR compared to CO_2. CFC-113 also absorbs IR in the atmospheric window. Hence, on a per molecule basis, CFC-113 exerts a larger positive radiative forcing compared to CO_2 (contributes more to global warming).

(Lecture 1 notes, Slide 53)
(Lecture 2a notes, Slide 16)

The number of carbon atoms present in the molecule is not the main factor. What matters is the wavelengths in which the molecules absorb, and their stability.

Please take note that there is, though, a higher concentration of CO_2 in the atmosphere; and hence overall, CO_2 remains the major greenhouse gas in the atmosphere.

(b) Coal with a high sulfur content burns to produce combustion products from carbon, but it also produces sulfate from burning of the sulfur. Sulfate emitted from smokestacks creates tropospheric sulfate clouds. Which two effects would emissions from this coal combustion have on global warming?

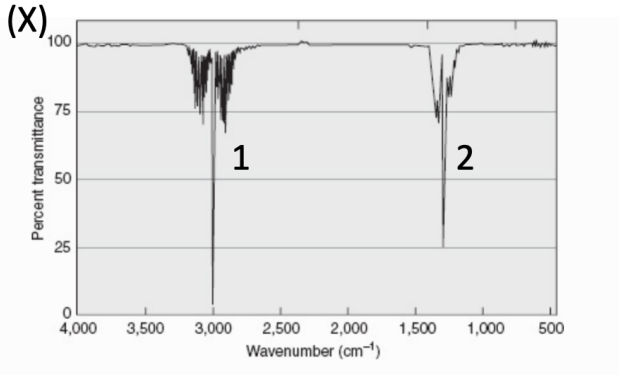
(3 marks)

Coal combustion produces CO_2, which would contribute to global warming.
Sulfate aerosols in the air would lead to cooling.

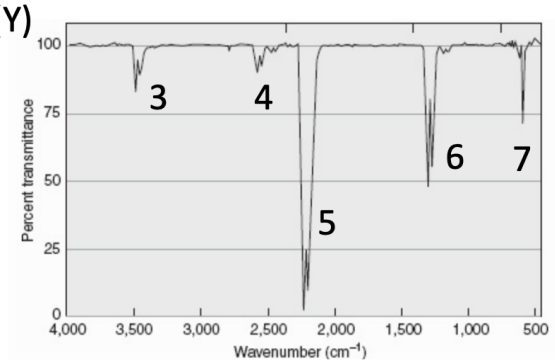
(Lecture 2a notes, Slide 19, 21)

(c) Below are the infrared absorption spectra of two Molecules (X) and (Y) that are currently being released into the atmosphere. These two molecules absorb infrared radiation at regions 1 to 7. Identify one absorption from each molecule that would contribute the most to the greenhouse effect. Rationalize your answer.

(X)



(Y)



(4 marks)

The absorptions that absorb in the atmospheric window between 8-12 um will cause greatest greenhouse effect. With this in mind, we can now convert the absorptions in cm^{-1} to μm.

Absorption 1 = 3000 cm^{-1} = 3.3 μm
Absorption 2 = 1350 cm^{-1} = 7.4 μm
Absorption 3 = 3400 cm^{-1} = 2.9 μm
Absorption 4 = 2600 cm^{-1} = 3.8 μm
Absorption 5= 2250 cm^{-1} = 4.4 μm
Absorption 6 = 1300 cm^{-1} = 7.7 μm
Absorption 7= 600 cm^{-1} = 16.7 μm

Absorptions 2 and 6 lie closest to the atmospheric window. Hence, they should contribute the most to the greenhouse effect.

(Lecture 2a notes, Slide 15)

Question 3

After graduating from NUS, you started working in the environmental protection agency of an urban city named Kroywen. Despite being a garden city, Kroywen suffers regularly from the photochemical smog. A local politician has requested your agency to measure the hydrocarbons, nitrogen oxides (NO_x), etc. in the air, so as to derive a pollutant concentration - time profile.

(a) Propose a method to quantify the hydrocarbons in the air.

(4 marks)

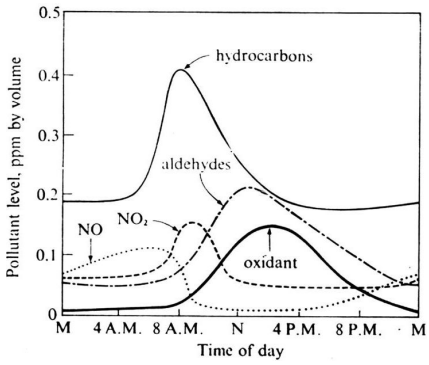
Hydrocarbons in the air can be analyzed using gas chromatography, with a field ionisation detector (FID).

Infrared spectroscopy alone would not work well, as there are many types of hydrocarbons, and their signals will likely overlap in the IR spectrum. Hence, it will be difficult to quantify them alone with IR.

It is also not very practical to combust the hydrocarbons and then quantify the CO_2. This is because there is an existing background level of CO_2 that still needs to be taken into account of.

(b) After much work, your agency produced the requested pollutant concentration - time profile data as shown on the right. On seeing it, the same local politician requested that all the hydrocarbons in the air should be eliminated, so as to eliminate all occurrences of the photochemical smog. The rationale for his strategy is that hydrocarbons have the highest concentration, hence their elimination will eliminate the photochemical smog. Comment on the feasibility of his strategy.

(6 marks)



The hydrocarbons in the air contribute to the formation of PAN and the regeneration of NO_2. Hence, a way to mitigate the photochemical smog in Kroywen is to lower the concentration of hydrocarbons. This can be done by ensuring that all the vehicles are mounted with catalytic convertors. However, there is a limit to what can be done to reduce the hydrocarbon level. This is because there will also be plants (as mentioned, Kroywen is a garden city) emitting the hydrocarbons. It is better to also reduce the amount of NO_2, which is the crucial molecule for tropospheric ozone formation.

(Lecture 3 notes, Slide 36)

Question 4

Singapore's electricity supply is provided mainly by gas turbines, which run on natural gas. These gas turbines emit 170 g of carbon for each kWh of energy produced. There are 1,100,000 households in Singapore. Each Singapore household uses an average of 5000 kWh of electric energy per year.

Note: One kilowatt-hour (1 kWh) is equal to the amount of energy you would use if you kept a single 1000 W appliance running for 1 hour.

(a) Calculate the amount of CO_2 (in tonnes) released to the atmosphere if all the electricity used by all the households in Singapore is generated by gas turbines.

(3 marks)

Amount of CO_2 released by one household = 5000 KWh $\times$ 170g of carbon per KWh $\times$ 44/12 $\times$ 10^{-6} = 3.1 tonne

Amount of CO_2 released by all households in Singapore = 3.1 tonne $\times$ 1,100,000 = 3.4 million tonnes of CO_2.

(b) Singapore aims to install on her shores a solar capacity of 2 gigawatt-peak by 2030. Assuming that all the solar panels installed could give this power for 8 hr a day (or 2920 hr a year), and the solar electricity generated is fed into the grid supplying electricity to the households in Singapore. If each Singapore household still uses an average of 5000 kWh of electric energy per year, will there still be a net emission of CO_2 by the households in Singapore? If so, how much more? List down assumptions (if any) made in your calculations.

(3 marks)

2 GW = 2 $\times$ 10^6 kW.
If solar panel operates for 2920 hrs, then the amount of electricity produced = 5,840 $\times$ 10^6 kWh.

The amount of electricity needed in Singapore = 5,000 kWh $\times$ 1,100,000 households = 5,500 $\times$ 10^6 kWh.

This indicates that the installation of solar panels with 2 GW_p capacity should produce enough electricity to power all the Singapore homes, with the assumption that electricity can be supplied on demand.

(c) In the context of Singapore and only considering the needs of the households, do you think it is practical to completely replace all its gas turbines with that of solar-generated electricity, and achieve zero CO_2 emissions? Rationalize your answer.

(4 marks)

No. The reason is that solar electricity is an intermittent source of electricity. A fuel that can be burnt in a generator is still needed to ensure that electricity can be supplied on demand. This can be provided by the use of a fuel such as H_2. However, the H_2 must be produced from a renewable source of energy, for example, water electrolysis powered by solar-electricity. Otherwise, there will be a net emission of CO_2.

(Lecture 4 & 5 notes, Slide 47)

Question 5

Petrol pricey? Japanese invent car that runs on water

By Reuters Life! 2 MIN READ

TOKYO (Reuters Life!) - Tired of petrol prices rising daily at the pump? A Japanese company has invented an electric-powered, and environmentally friendly, car that it says runs solely on water.

Genepax unveiled the car in the western city of Osaka on Thursday, saying that a liter (2.1 pints) of any kind of water -- rain, river or sea -- was all you needed to get the engine going for about an hour at a speed of 80 km (50 miles).

"The car will continue to run as long as you have a bottle of water to top up from time to time," Genepax CEO Kiyoshi Hirasawa told local broadcaster TV Tokyo.

"It does not require you to build up an infrastructure to recharge your batteries, which is usually the case for most electric cars," he added.

Once the water is poured into the tank at the back of the car, the a generator breaks it down and uses it to create electrical power, TV Tokyo said.

https://www.reuters.com/article/us-japan-car-water-idINSP7366720080613

You found the above report from the Reuters newspaper website. It describes a car that is powered solely by water. Explain, using scientific principles, how this 'water-powered' car might be like, and whether it actually works or not.

(10 marks)

The water-powered car will not work. This is because water is not a fuel.
Even if the generator (possibly an electrolyzer) breaks water down into H_2, and the H_2 is converted to electricity via a fuel cell, there still needs to be an original source of power. Hence, it is not possible to have a car that is powered solely on water. Most big automakers are working on fuel-cell cars that run on hydrogen and emit -- not consume -- water.

(Lecture 4 & 5 notes, Slide 45)

You may continue your answer for Q5 here…

End-of-paper

CM5244 Advanced Environmental Chemistry (Semester 2, 2023/2024)

Test (18th April 2024)

Examiner: Assoc. Prof. Boon Siang Jason YEO

Time allowed: 1 h, 15 min

Student No: _____

INSTRUCTIONS

1. Please write your student number only. Do not write your name.
2. Please answer all the questions by yourself. Communication with the others is forbidden.
3. Write your answers in the space provided below each question.
4. You may consult notes / textbooks (only hard copies).
5. You may use a calculator. However, electronic devices such as laptops / mobile phones / tablets / ipads are not allowed.
6. Just like what you have done for the assignments, you need to make use of your creativity to answer the questions.
7. There are 10 printed pages in this script.

Question 1

(a) HCFC-123 (HCFC = hydrochlorofluorocarbon) has been proposed as a substitute for CFC-11 in low pressure refrigeration. How would the following environmental properties differ between these two chemicals?

(i) Tropospheric lifetime
(ii) Ozone-depleting potential (ODP)

Please give reasons for your answers.

(5 marks)

HCFC-123 = $C_2HF_3Cl_2$
(Use of Rule of 90 in Lecture Notes 1, Slide 53 to determine the molecular formula of HCFC-123 = 123+90 =213)

CFC-11 = $CFCl_3$

(a) The tropospheric lifetime of HCFC-123 is shorter, as it has an abstractable H, unlike CFC-11.
(b) HFC-123 has a smaller ODP than CFC-11, as it has a shorter lifetime. Moreover, HFC-123 a has lesser number of chlorine atoms.

(b) CFC-114 has a tropospheric lifetime of 236 years. Would you expect CFC-113 and CFC-115 to have shorter or longer lifetimes compared to CFC-114? How about their relative global warming potentials (GWP)? Please give reasons for your answers.

(5 marks)

CFC-113 = $C_2F_3Cl_3$
CFC-114 = $C_2F_4Cl_2$
CFC-115 = C_2F_5Cl

We expect CFC-115 to have a longer lifetime than CFC-114, because it has only 1 C-Cl bond, which is the weakest bond in the molecule. In contrast, CFC-114 has two C-Cl bonds, which makes it more susceptible to degradation.

We expect CFC-115 to have a larger GWP, compared to CFC-114, because of its longer lifetime.

CFC-113 would have both the smallest lifetime and GWP.

(c) What is/are the expected product(s) when ClO• reacts with NO•? What are the possible fates of the product(s) of this reaction, if it takes place in the lower stratosphere? Devise a mechanism incorporating all these reactions. Does the net result of this mechanism indicate the sustained destruction of stratospheric ozone?

(5 marks)

$ClO\bullet + NO\bullet \rightarrow Cl\bullet + NO_2$

$NO_2 \rightarrow NO\bullet + O\bullet$ (this equation shows the fate of NO_2)

$Cl\bullet + O_3 \rightarrow ClO\bullet + O_2$ (this equation shows the fate of Cl•)

$O_3 \rightarrow O_2 + O\bullet$

Yes, ozone can be sustainably destroyed in this mechanism.

Question 2

(a) Explain why HCFC-22 has more infrared (IR) absorption bands with a greater range of IR wavelengths absorbed, and absorbs IR much more efficiently on a per molecule basis, than CH_4, the hydrocarbon from which it is derived.

(5 marks)

HCFC-22 = CHF_2Cl vs. CH_4

HCFC-22 is less symmetrical than CH_4, so it would have more IR absorption bands. Moreover, its bonds (C-F, C-Cl) are also more polar than those of CH_4. Hence, it would absorb IR more efficiently on a per molecule basis, as compared to CH_4.

(b) Which of the following molecule(s) in the table below has/have the potential to cause a greenhouse effect similar to CFCs? If you have identified polyatomic molecule(s), what are the specific vibration(s) involved?

Molecule	S-C-O (linear)	H-Cl	F-F	H_2S (non-linear)	Cl_2O (non-linear)
Vibrations	11,641 nm 18,975 nm 4810 nm	3465 nm	11,211 nm	3830 nm 7752 nm 3726 nm	14,706 nm 30,303 nm 10,277 nm

(5 marks)

CFCs absorb in the atmospheric window of 8000-12000 nm. We need to find molecules with IR active vibrations within this region (Lecture Notes 2a, Slide 15).

S-C-O absorbs at 11,641 nm.
Cl_2O absorbs at 10,277 nm (Note that all 3 vibrational modes are IR-active).

These two above-mentioned molecules should behave like CFCs.

F_2 is not IR-active (Lecture Notes 2a, Slide 12)

Question 3

(a) After graduating from the NUS, you started working in the environmental protection agency of a modern city located near a major shipping port. Your supervisor has tasked you to measure the indoor and outdoor concentrations of carbon monoxide (CO), carbon dioxide (CO_2) and sulfur dioxide (SO_2). For the indoor measurements, you will be going to the homes of several volunteers. After some hard work, you discover that the (i) indoor/outdoor concentration ratios for CO are usually near 1. Could you predict what might be the indoor/outdoor concentration ratios for (ii) CO_2 and for (iii) SO_2?

Rationalize your answers.

(5 marks)

CO is an inert gas. This might explain why its indoor and outdoor concentrations are similar. Humans breath out CO_2. In the outdoors, the CO_2 could be more easily dispersed, or plants can also absorb the CO_2. Hence the indoor/outdoor concentration of CO_2 >1. SO_2 is mainly produced by the burning of ship bunker fuel, which tend to be high in sulfur. Hence the indoor/outdoor concentration of SO_2 <1.

(b) Propose a method to quantify the concentration of CO_2 in the air. Rationalize your chosen approach.

(5 marks)

CO_2 could be measured using gas chromatography or Infrared spectroscopy. IR spectroscopy works because CO_2 has IR-active vibrational modes. The Beer's Law is used to quantify the concentrations of CO_2.

Question 4

'Trees absorb CO_2 during photosynthesis. Hence, we shall completely eliminate air pollution around our industrial incinerators by planting a pine and citrus trees greenbelt around it.'

'Our cars are solely powered by ammonia, NH_3 (in ammonia fuel cells), and would thus only produce H_2O and N_2 emissions. Therefore, their usage must be totally environmentally friendly.'

I recall reading the above two statements in a magazine. They seem reasonably correct. But if you think that they might be (partially) wrong, please justify your position.

(10 marks)

Pines and citrus trees emit olefins. These olefins can play a role in generating the photochemical smog (See Lecture Notes 3, Slide 17).

This depends primarily on how the ammonia is formed. If the ammonia is formed (Haber-Bosch process) using H_2 that are produced via steam-reforming of methane, then the process lead to CO_2 emissions (Lecture Notes 4 and 5, Slides 9 and 54) and cannot be considered environmentally friendly. Other considerations include the product of NH_3 oxidation, which might be NO_x.

You may use the next page to answer Q4

You may use this page to answer Q4

Question 5

Singapore's total electricity consumption is 54.9 TW.h per year. Singapore's electricity supply is provided mainly by burning fossil fuels. However, it has expressed a strong desire to replace its current electricity production method with renewables.

(a) One method that has been thought of is to use solar panels only for electricity production. Is this practical? Justify your response. Assume that a solar panel is 1.7 m^2 in surface area. It is also able to give a maximum power output of 330 W. The total land area of Singapore is 734 km^2, of which most of it is urbanized.

List down other assumptions (if any) made in your calculations.

(5 marks)

We assume that the solar panels are able to give a maximum power output for 4 hrs a day (Singapore can get very cloudy and rainy for some days).

Number of solar panels needed to provide 54.9 TW.h electricity per year = 54.9 TW.h / (330 W × 4 hrs × 365 days) = 54.9×10^{12} / (330 W × 4 hrs × 365 days) = 1.1×10^8 panels.

Area needed to lay out panels = 1.1×10^8 panels × (1.7×10^{-6} km^2) = 1.9×10^2 km^2

Singapore would have use ¼ of its land for solar panels! This is not very practical, considering how urbanized it is.

Furthermore, electricity produced using solar panels are intermittent in nature (Lecture Notes 4 & 5, Slides 49 and 50)!

(b) Another method is based on 3rd generation biofuels, which focuses on extracting resources from microalgae. How practical is this approach? Justify your response. Given that specialized microalgae strains that grow in the sea with 2 % photosynthesis efficiencies (solar-to-biomass) have been cultivated. Singapore has a total coastline of 228 km, and that it receives an annual solar radiation of 1620 KW.h/m^2.

List down other assumptions (if any) made in your calculations.

(5 marks)

Let the total sea area needed to grow the microalgae be Z km^2. Assume that the biomass to electricity conversion efficiency is 50 %.

54.9 TW.h electricity per year = 2% × 1620 KW.h/m^2 × (Z) × 50%

54.9×10^{12} W.h electricity per year = 2/100 × 1620 KW.h/ 10^{-6} km^2 × (Z) × 50/100

Z = 3.4×10^6 km^2

The breadth is = 3.4×10^6 km^2 / 228 km = 14 km breadth.

This does not appear to be possible, as Singapore would need to grow microalgae that stretches 14 km off shore!

(c) In the context of Singapore, what do you think is the main constraint that prevents it from producing more renewable electricity?

(5 marks)

Scarcity of Land is the major constraint.

(d) What do you think are the strategies which Singapore could adopt in order to reduce its dependence on fossil fuels?

(5 marks)

No single strategy will work for Singapore! It is necessary to tap into as many forms of renewables (solar, geothermal, hydro) as possible, while trying to reduce power consumption. Import electricity from our neighboring countries, etc.

End-of-paper

CM5244 Advanced Environmental Chemistry (Semester 2, 2024/2025)

Test (10th April 2025)

Examiner: Assoc. Prof. Boon Siang Jason YEO

Student No: _____

INSTRUCTIONS

1. Please write your student number only. Do not write your name.
2. Please answer all the questions by yourself.
3. Write your answers in the space provided below each question.
4. You may consult notes / textbooks (only hard copies).
5. You may use a calculator. However, electronic devices such as laptops / mobile phones / tablets / ipads are not allowed.
6. Just like what you have done for the assignments, you need to make use of your creativity to answer the questions. **The most important aspect to answering some of the questions is to be able to rationalize your answer.**
7. There are 10 printed pages in this script.

Answers will be uploaded later

Question 1

(a) HFC-245 (HFC = hydrofluorocarbon) has been proposed as a substitute for CFC-11 in low pressure refrigeration. To assess this feasibility, you would need to determine its possible effect on the stratospheric ozone layer, its global warming potential (GWP) and its role as a component in the photochemical smog.

How would the following two environmental properties differ between these two chemicals?

(i) Tropospheric lifetime
(ii) Ozone-depleting potential (ODP)

Please give reasons for your answers.

Note: The Ozone Depletion Potential (ODP) measures a chemical's relative ability to deplete the ozone layer compared to CFC-11, which has an ODP of 1.0.)

(5 marks)

(b) Contrary to what many people may think, HFC-245 actually has a smaller global warming potential (GWP) compared to CFC-11. Could you explain why?

(5 marks)

(c) Could you try predicting whether HFC-245 or CFC-11 would play a larger role in the formation of a photochemical smog? Give your reasons.

(5 marks)

(d) Based on your analyses made in (a)-(c), would you choose HFC-245 over CFC-11 for low pressure refrigeration? Give your reasons.

(5 marks)

Question 2

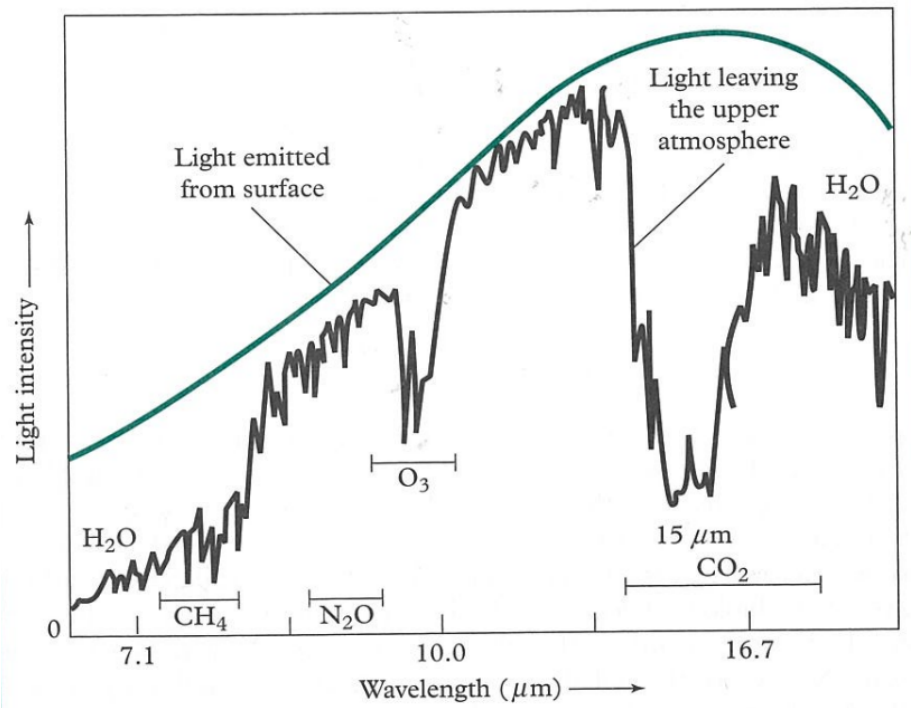
Two common tropospheric pollutant gases are SO_2 and NO_2. The wavelengths for their vibrations are given in the table below.

Molecule	Symmetric stretch	Antisymmetric stretch	Bending
SO_2	8.7 µm	7.3 µm	19.3 µm
NO_2	7.6 µm	6.2 µm	13.3 µm

(a) Which of the above vibrations are capable of absorbing infrared energy?

(5 marks)

(b) Based on the wavelengths of the IR-absorbing vibrations of NO_2 and SO_2, and the spectra shown below, decide which, if any, of the vibrations of NO_2 and SO_2 could contribute **<u>much</u>** to the greenhouse effect.



The figure above shows '*<u>experimentally measured intensity of thermal IR light leaving the earth's upper atmosphere</u>*', compared with the '*<u>theoretical intensity that would be expected without absorption by atmospheric greenhouse gases</u>*'.

(5 marks)

(c) Do you think NO_2 and SO_2 play major roles in global warming? Give your reasons.

(5 marks)

(d) CO_2 and N_2O are also greenhouse gases. Apart from using infrared spectroscopy, what technique(s) could you used to determine their concentrations in the atmosphere?

(5 marks)

(e) The *Kyoto Protocol*, an international agreement adopted in 1997, aimed to reduce CO_2 and other greenhouse gases (GHGs) emissions by requiring industrialized nations to set binding emission reduction targets. In reality, H_2O vapor (average concentration of 0.4 %) is a dominant greenhouse gas. Yet unlike CO_2 and many other GHGs, its emission is not regulated. Why not?

(5 marks)

Question 3

'Carbon monoxide is dangerous to human health, because it is an important ingredient of the photochemical smog.'

'Biomass fuels cannot be considered at all as a renewable and sustainable source of fuel. This is because when combusted, such fuels would produce a large amount of CO_2 per unit of heat produced.'

'The natural formation and destruction of tropospheric ozone can be explained by the Chapman Cycle'.

I recall reading the above three statements in a magazine. Do you agree with them? Please justify your position.

(10 marks)

This page can be used to answer Question 3

Question 4

After graduating from the NUS, you worked for a government agency in Singapore. Your work is to discover new technologies that could lead to the replacement of conventional fossil jet fuel with biofuels. A company recently came to your agency with a claim that they have developed a crop that has a state-of-the-art 10% efficiency of converting solar-to-biomass fuel, and they could replace all fossil jet fuel with their own biomass fuel. Your boss would like you to examine how feasible it is to adopt this new technology in Singapore. Write a brief report to advise your boss on the feasibility of this claim.

This is the data that you collected from various sources:

- A commercial Boeing-777 requires 7 tonnes of jet fuel for each hour of flight. The gravimetric energy density of fossil jet fuel is 43 MJ/kg.

- An average of 500 flights depart from Singapore each day, and the average flight time of each flight is 7 hrs.

- Singapore's total land mass is 734 km^2, almost all of which is urbanized. Realistically, only some offshore islands like Sentosa, Pulau Ubin, etc. could be used for growing crops. These have a total land size of 5.6 km^2.

- Singapore receives an annual solar radiation of 1620 $KW.h/m^2$.

List down other assumptions (if any) made in your assessment

(10 marks)

This page can be used to answer Question 4

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